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MASS AND WEIGHT





MASS AND WEIGHT



- **Units of Measurement:**

- **Kilogram (kg):** Used to measure heavier objects, e.g., a bag of sugar.



- **Gram (g):** Used to measure lighter objects, e.g., a paperclip.



- **Exploring Different Types of Balances and Scales.**

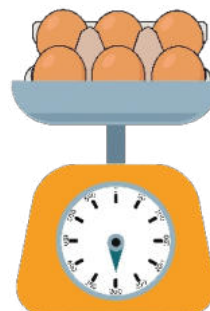
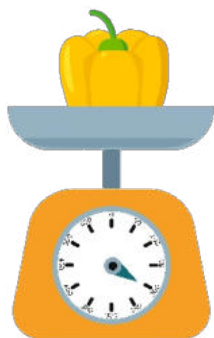
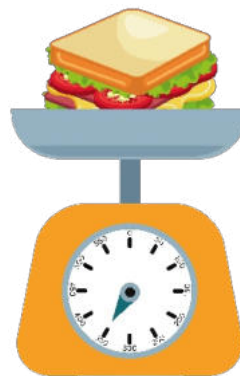
1. Common Balance: This balance has two pans. Place the item on one pan and weights on the other to find the item's weight.



2. Weighing Machine: This machine shows the weight directly on a dial without using separate weights.



- **Look at the weighing scales. Write the mass of each object.**



- **Conversion:**

$$1 \text{ kg} = 1,000 \text{ g}$$

- **Conversion of Units**

- **Convert Grams to Kilograms**

- **Formula:**

$$\text{Grams} \div 1000 = \text{kg}$$

- **Example:**

Convert 5,000 g to kg:

$$5,000 \text{ g} = 5,000 \div 1,000 = 5 \text{ kg}$$

- **Practice Problems.**

1) **3,500** g to kg = _____

2) **7,250** g to kg = _____

3) **12,000** g to kg = _____

4) **15,500** g to kg = _____

5) **20,000** g to kg = _____

6) **9,800** g to kg = _____



7) **4,600 g** to kg = _____

8) **18,750 g** to kg = _____

9) **2,500 g** to kg = _____

10) **11,250 g** to kg = _____

11) **25,000 g** to kg = _____

12) **14,750 g** to kg = _____

- **Convert Kilograms to Grams**

- **Formula:**

Weight in g = Weight in kg $\times$ 1,000

- **Example:**

Convert 34 kg 108 g to g:

$$(34 \times 1,000) \text{ g} + 108 \text{ g} = 34,000 \text{ g} + 108 \text{ g} = 34,108 \text{ g}$$

- **Practice Problems.**

1) **2 kg 500 g** to g = _____



2) **7 kg 120 g** to g = _____

3) **15 kg 80 g** to g = _____

4) **9 kg 450 g** to g = _____

5) **10 kg 250 g** to g = _____

6) **4 kg 980 g** to g = _____

7) **3 kg 725 g** to g = _____

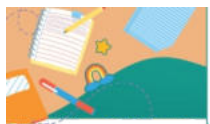
8) **11 kg 175 g** to g = _____

9) **18 kg 600 g** to g = _____

10) **25 kg 320 g** to g = _____

11) **6 kg 540 g** to g = _____

12) **14 kg 680 g** to g = _____



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• Addition of Mass

Steps:

1. Write kilograms (kg) and grams (g) in separate columns.
2. Add grams first. If grams exceed 1,000, carry over to kilograms.

• Example Problems.

1) $12 \text{ kg } 470 \text{ g} + 6 \text{ kg } 234 \text{ g} = \underline{\hspace{2cm}}$

2) $62 \text{ kg } 670 \text{ g} + 72 \text{ kg } 143 \text{ g} = \underline{\hspace{2cm}}$

3) $23 \text{ kg } 650 \text{ g} + 54 \text{ kg } 137 \text{ g} = \underline{\hspace{2cm}}$

4) $2 \text{ kg } 143 \text{ g} + 6 \text{ kg } 156 \text{ g} = \underline{\hspace{2cm}}$

5) $18 \text{ kg } 925 \text{ g} + 12 \text{ kg } 575 \text{ g} = \underline{\hspace{2cm}}$

6) $35 \text{ kg } 460 \text{ g} + 27 \text{ kg } 340 \text{ g} = \underline{\hspace{2cm}}$

7) $40 \text{ kg } 725 \text{ g} + 32 \text{ kg } 175 \text{ g} = \underline{\hspace{2cm}}$

8) $14 \text{ kg } 810 \text{ g} + 9 \text{ kg } 190 \text{ g} = \underline{\hspace{2cm}}$



9) **25 kg 900 g + 18 kg 650 g = _____**

10) **7 kg 250 g + 37 kg 800 g = _____**

11) **50 kg 430 g + 37 kg 870 g = _____**

12) **22 kg 540 g + 18 kg 360 g = _____**

● Subtraction of Mass

Steps:

1. Write kilograms (kg) and grams (g) in separate columns.
2. If grams in the top row are smaller, borrow 1 kg (1 kg = **1,000 g**).



● Example Problems.

1) **9 kg 210 g - 2 kg 100 g = _____**

2) **5 kg 700 g - 4 kg 450 g = _____**

3) **8 kg 850 g - 8 kg 700 g = _____**

4) **3 kg 900 g - 2 kg 160 g = _____**

5) **15 kg 400 g - 7 kg 250 g = _____**

6) **12 kg 580 g - 9 kg 375 g = _____**

7) **25 kg 650 g - 18 kg 470 g = _____**

8) **40 kg 725 g - 27 kg 340 g = _____**

9) **18 kg 450 g - 9 kg 600 g = _____**

10) **30 kg 800 g - 15 kg 425 g = _____**



11) **7 kg 250 g - 5 kg 800 g = _____**

12) **22 kg 540 g - 18 kg 360 g = _____**

• Weight Comparisons

Compare the following weights using **>**, **<**, **=**:

1) **250 g** **2 kg**

6) **1,500 g** **1.5 kg**

2) **7,000 g** **7 kg**

7) **800 g** **1 kg**

3) **20 kg** **250 g**

8) **2 kg** **675 g**

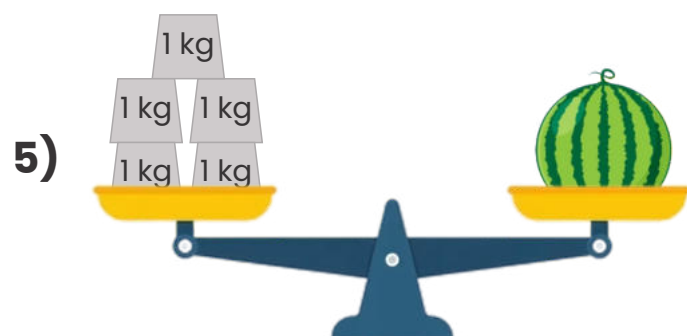
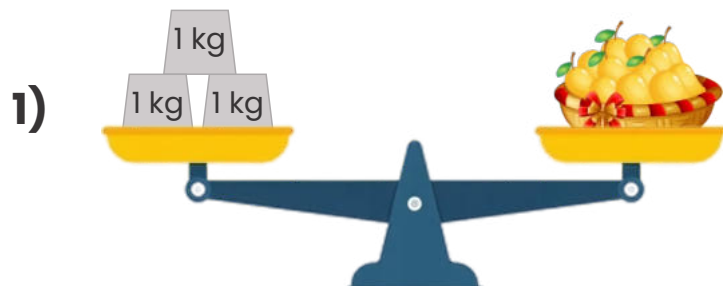
4) **7,500 g** **7 kg**

9) **3,000 g** **1.5 kg**

5) **35 g** **3 kg**

10) **9 kg** **6,000 g**

- Measure the weight of the objects.



- **Convert Kilograms to Grams.**

1) 2 kg to g = _____

2) 100 g to kg = _____

3) 3 kg to g = _____

4) 50 g to kg = _____

- **Word Problems**

Q1: John purchases 7 kg 200 g of sugar and 9 kg 395 g of rice. What is the total weight?





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Q2: Raheela buys 8 kg 400 g of tomatoes, 2 kg 130 g of brinjal, and 7 kg 300 g of watermelon. How much does she buy in total?



Q3: The total weight of Tania's bag is 45 kg 750 g, and Diana's bag is 43 kg 950 g. Whose bag is heavier and by how much?





Q4: A truck carries 352 kg 100 g of pumpkins and 207 kg 432 g of watermelons. Find the total weight.



Q5: A shopkeeper sells 67 kg 626 g of wheat on Saturday and 125 kg 200 g on Sunday. Find the total weight of wheat sold.

